AI COMPUTER VISION PROJECT

People Identification via Face Recognition

Project Team:

Z. Muhammad I M. Islam
M. Abdul Gafoor I H. Eltahir

Instructor:

Dr. Dwarikanath Mahapatra
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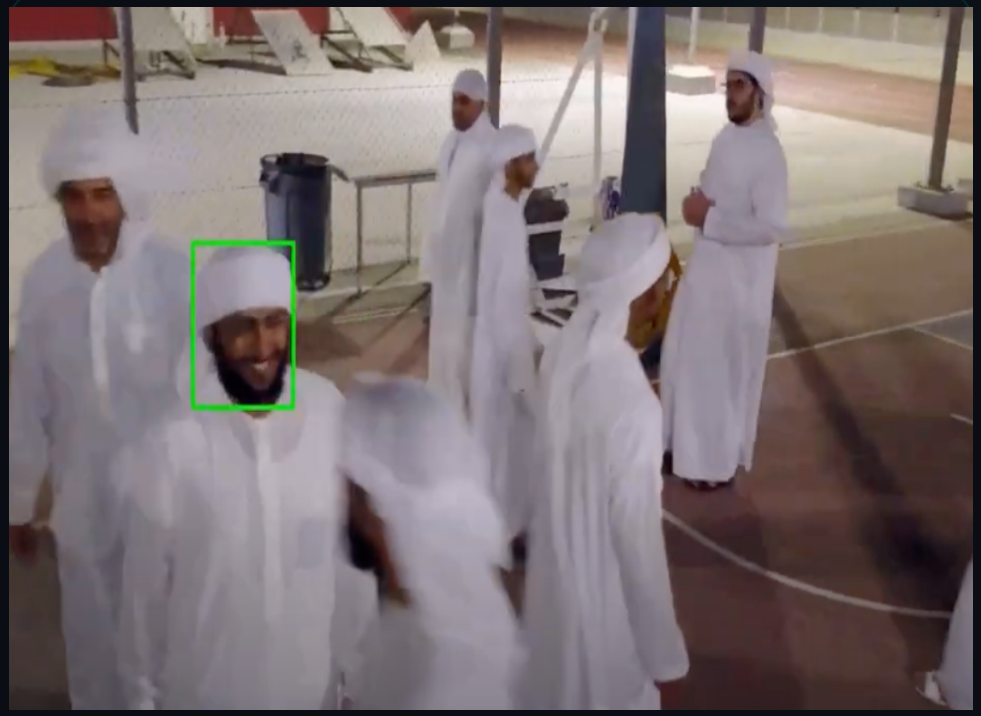
PROJECT OVERVIEW

The Challenge

Developing a robust system for human identification in the wild. Our project will deal with these objectives:

1. Detect all faces in the image.
2. Extract face signatures from the detected faces
3. Select the face that corresponds to the minimum distance



THE DRONEFACE DATASET

Multi-Source Composition

We utilized three distinct data directories to build and test our verification architecture:

Gallery (1,364 images): 11 identities (A-K), pre-cropped at 124 images each for high-density training.

Scenes (620 raw): Outdoor/Indoor captures with 3,680x2,760 resolution & ultra-wide FOV.

Portraits: 3-direction baseline shots used for ground-truth calibration.

END-TO-END PIPELINE

1. Input Stage

Raw scene processing and MTCNN based detection. Faces validated at a 10x10 minimum pixel resolution.

2. Processing

Feature extraction using Facenet512. Generates high dimensional vector embeddings (signatures).

3. Decision

Voting based gallery matcher applies 4 layers of distance gates to confirm or reject identities.

GALLERY CONSTRUCTION

Embedding Extraction

Using **FaceNet** as the primary signature model. It produces consistent cosine distances: ~0.37 for same-person matches and above ~0.37 + for different individuals.

Vector Search

A DataFrame stores 1,364 float32 NumPy arrays. Each probe face is matched against this global gallery in real-time during scene processing.

VOTING MATCHER LOGIC

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Distance Gates

The Decision Engine

To maximize precision, we implemented a cascaded filtering mechanism:

Match Threshold: 0.55 (Cosine Distance max)

Vote Floor: Minimum 20 rows must match.

Mean Gate: Avg match distance < 0.48.

Best Gate: Min match distance < 0.37.

COSINE SIMILARITY CORE

$$d(a,b) = 1 - \frac{a \cdot b}{\|a\| \, \|b\|}$$

METRIC DEFINITIONS

Metric	Definition	Project Focus
Precision	TP / (TP + FP)	Reducing spurious identifications (Conservative).
Recall	TP / lExpectedl	Recovering all identities in blurry scenes.
Token_OK	Strict Exact Match	Scenes with zero FP and zero FN (Exact hit).
Rejected	Distance Filtered	Faces below threshold are ignored (No FP penalty).

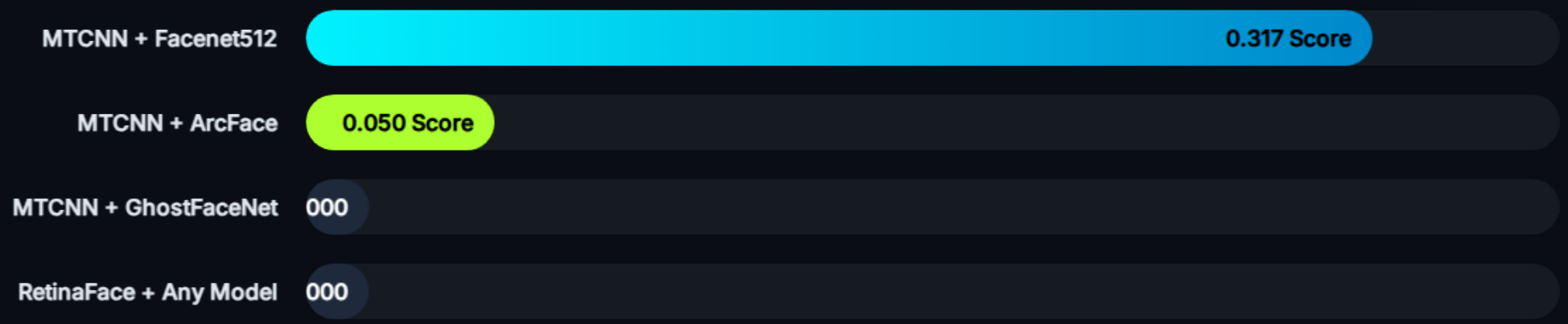
METRIC Examples

- TP / FP / FN: Example: expected {C, D}, predicted {C, D, J}. TP = 2 because C and D are correct, FP = 1 because J is extra, FN = 0 because nothing was missed.

- token_ok / Score: token_ok is true only for an exact match. Example: expected {C, D}, predicted {C, D} → token_ok = True; predicted {C, D, J} → False.

- Grid-search Score = 0.5 × mean_precision + 0.5 × mean_recall; example: 0.5 × 0.3667 + 0.5 × 0.2667 = 0.3167.

OPTIMIZATION GRID SEARCH



WINNER: MTCNN + Facenet512 at Threshold 0.55

Score = 0.5 × mean_precision + 0.5 × mean_recall

Only tested for 15-20 scenes, hence low scores for some

CASE STUDY

Scene Analysis: cd_gp_0_eo_12

Correct Identification: Identities C and D successfully matched despite variable angles.

Rejection Success: Face 0 (A candidate) rejected via distance gate (Avg 0.49 > 0.48).

Failure Mode: Identity J triggered a False Positive due to background noise.

BASELINE RESULTS (620 SCENES)

- **77.85% Mean Precision:** System is highly reliable when making a prediction.
- **59.01% Mean Recall:** Conservative gates often miss low-quality faces.
- **40.0% Token OK:** 248/620 scenes were identified with 100% perfection.

Note: Baseline uses MTCNN Detector + Facenet512 Recognizer.

PERFORMANCE BOTTLENECKS

- **Zero Fine-Tuning:** Pre-trained models (VGGFace2/MS-Celeb) were not optimized for the 11 specific identities (A-K).
- **Image Quality:** Drone-simulated blur loses fine-grained texture data, causing embedding proximity collapse.
- **Demographic Mismatch:** Training sets biased toward Western faces underperform on the project's East Asian subjects.
- **Detection Failure:** RetinaFace failed to generate sufficient proposals in low-quality scenes.

Conclusion & Future

The project validated that **MTCNN + Facenet512** with custom **cascaded voting gates** provides a robust, conservative baseline for person identification.

Next Step: Super-Resolution

Next Step: Domain Fine-Tuning

THANK YOU FOR YOUR ATTENTION
Q&A SESSION